

Vital Signs Monitoring in Various Conditions Using the Single Antenna Bio-Sensing Method

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Abstract—Continuous monitoring of respiration is essential, as it serves as a key biomarker for health assessment. It has a wide range of applications, from medical to industrial settings, including the early detection of severe medical conditions, management of chronic illnesses, and evaluation of worker fatigue in high-risk environments. In this context, Single Antenna Bio-Sensing (SABioS) offers a promising new approach to respiratory monitoring, emphasizing comfort and mobility by removing the need for skin contact, tight-fitting devices, or external readers. This paper investigates the capabilities of SABioS through a series of experiments that assess its effectiveness across different placements, body positions, and distances from the chest. The study also examines the influence of operational frequency using various antenna designs. By analyzing the signal-to-noise ratio (SNR) in diverse scenarios, this research provides valuable insights into the robustness and adaptability of SABioS, paving the way for its development as a practical, non-invasive respiratory monitoring solution.

Index Terms—Respiratory Monitoring, Cardiac Monitoring, Vital Signs, Antenna Sensors, Non-invasive sensing, Near-field sensing, Biosensor, Industrial health monitoring, Wearables.

I. INTRODUCTION

Respiration is a crucial vital sign for assessing a patient's overall health and predicting severe medical events, such as intensive care unit (ICU) admissions, respiratory failure, and cardiac arrest [1]. Continuous monitoring of this biosignal is also essential for managing and evaluating the progression of chronic conditions, including sleep apnea and asthma [2].

Beyond clinical applications, monitoring respiratory activity is also valuable in industrial settings, where it can help assess the health conditions of workers in remote locations, such as in mines, maritime, or aviation environments. Additionally, it plays a crucial role in evaluating fatigue and stress levels in high-risk occupations or critical roles [3], potentially reducing workplace accidents and enabling rapid emergency responses.

Despite the critical role of respiratory monitoring in industrial safety, medical applications, athletic performance track-

ing, and more, state-of-the-art portable systems capable of continuous monitoring often rely on invasive methods, such as nasal cannulae or mouthpieces [4], or require tightly fastened chest belts that use resistive [5], inductive [6], capacitive [7], or optical fiber [8] sensing elements. Some newer variants embed sensors into tight-fitting shirts, sacrificing accuracy for comfort, while others utilize piezoelectric sensors [9], which must be applied directly to the skin, potentially causing discomfort and irritation.

These limitations hinder the applicability of current respiratory monitoring technologies, as they can interfere with daily activities and even disrupt the natural respiratory cycle of users, whether they are workers in high-risk roles or patients with medical conditions. Recent research has explored novel approaches, such as employing on-body antennas for respiratory detection. However, these methods often rely on technologies like Bluetooth RSSI monitoring [10], [11] or RFID tag reading [12], which typically require an external adjacent device to function, limiting mobility and introducing additional sources of error and instability due to interference and signal path variances between the two parties.

Recently, Single Antenna Bio-Sensing (SABioS) method is introduced [13], which utilizes antenna near fields to monitor respiratory and also cardiac activity, with no need for external readers or base stations, enabling local installation on the subject. The standalone operation and employing near fields allows SABioS to offer superior signal quality compared to other state-of-the-art antenna-based respiratory monitoring systems. It does not require a tight fit or direct skin contact, making it highly portable and suitable for sensitive medical and industrial applications. As a relatively new approach, SABioS still requires further examination to fully understand its potential and limitations across different configurations.

In this paper, a series of experiments are conducted to evaluate the robustness of the previously-introduced SABioS method [13] in capturing vital signs across an extended set of configurations and conditions. Samples from each experiment are provided, and the Signal-to-Noise Ratio (SNR) is calculated for each scenario to facilitate a better comparison. The

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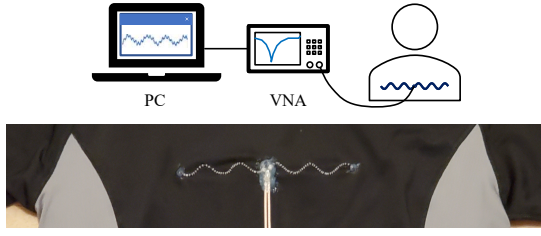


Fig. 1. Top: Diagram of the experimental setup: A PC records data captured by a VNA device from an antenna positioned on the subject. Bottom: Prototype of a 50 Ω Sinusoidal dipole antenna installed over a T-shirt chest.

paper begins by revisiting the operating principle of SABioS. Next, the impact of sensor placement on different body regions is explored, followed by an assessment of its performance in recumbent positions. The performance of the antenna in detecting vital signs is then evaluated when there is a distance between the antenna and the chest. Finally, the effect of the operating frequency is examined using an extended set of antennas to demonstrate the robustness of SABioS in vital sign detection under varying conditions.

II. ANTENNA SENSORS FOR VITAL SIGNS MONITORING

Antennas can be used to monitor vital signs by detecting changes in the dielectric properties and structural movements of the thorax and abdomen [13]. Respiration involves the contraction and relaxation of the diaphragm and intercostal muscles, resulting in variations in lung volume and intra-thoracic pressure, and ultimately expanding the circumference of the thorax. This expansion during respiration correlates with tidal volume and can serve as an indicator of different phases of the breathing cycle [14], [15].

The movement of internal organs not only affects the body's circumference but also alters its electromagnetic properties. For instance, at a frequency of 915 MHz, the lungs undergo significant changes in relative permittivity, decreasing from approximately 51.4 when deflated to around 22 when filled with air [16]. This sharp variation influences the antenna's input impedance when placed near the thorax [17], enabling the antenna to detect internal organ movements without the need for a tight chest band or direct skin contact. However, as an alternative contributor, if the antenna is made from flexible materials, it can also detect chest expansions during respiratory motion, which causes structural deformation and consequently modifies its input impedance [18], [19].

Equation (1) presents the reflection coefficient (S_{11}) of the antenna, where Z_{ant} is the input impedance of the antenna sensor, and Z_0 is the characteristic impedance of the interrogation system [20]. By analyzing changes in the S_{11} due to impedance shifts, vital signs can be tracked [13].

$$S_{11} = \frac{Z_{ant} - Z_0}{Z_{ant} + Z_0} \quad (1)$$

Antennas generally exhibit strong near fields [17], making their input impedance highly sensitive to changes in permittivity of materials in their proximity. Moreover, since

antenna characteristics are influenced by geometry, stretchable antennas can also respond significantly to deformations. For instance, a sinusoidal dipole antenna made of flexible material can be used as a wearable sensor to detect permittivity shifts and/or chest expansions [13]. Other antenna types, such as variants of monopoles, loop antennas [21], and microstrip patch antennas [17], can also be employed for this application. Additionally, these antennas can be worn in various configurations, over different regions of the upper body, and under different conditions. However, these diverse configurations need to be studied and tested to assess their applicability.

To test these different scenarios, the setup shown in Fig. 1 was prepared, consisting of a PC recording data from a Vector Network Analyzer (VNA) acting as the detector, which is connected to a on-body sinusoidal dipole antenna made of flexible conducting polymers [18]. The designed antenna resonates at 827 MHz in free space and around 808 MHz when installed on the body. The sampling rate of the VNA is 230 Hz, which is sufficient for capturing vital signs. The experimental protocol for SABioS has been approved by the Ethics Review Board of CHU de Québec – Université Laval Hospital.

A. Antenna Placement Effect

The antenna sensor can be installed on the upper body in various configurations. It can be placed on the front side, spanning from the upper chest to the abdomen. Alternatively, it can be positioned on the upper back, which is similar to cases where the antenna is mounted on the back of a chair. Another option is to wear it diagonally across the front of the chest, resembling the placement of an antenna on the strap of a cross-shoulder bag or a vehicle's seat belt.

Fig. 2 presents the sample signals recorded when the antenna is placed at various locations on the body, including both the raw and the extracted respiratory signal, which was filtered using a simple low-pass filter with a cutoff at 0.66 Hz (corresponding to 40 bpm). It is clear that the wearing options that capture chest expansion provide the best signal-to-noise ratio for respiration. For instance, when the sensor is embedded on a cross-shoulder strap, respiratory signal capture is weakened.

Additionally, the harmonics related to cardiac activity are most effectively captured when the antenna is placed directly over the heart or at the front of the body, such as in the middle of the chest or in the cross-shoulder configuration. This suggests that, for applications requiring cardiac activity monitoring, it is essential to place the sensor at the center of the chest to ensure the antenna's near fields reach the heart, providing the maximum response to heart movements.

B. Monitoring in Recumbent Positions

The next set of experiments presented in Fig. 3 assess the feasibility of using the antenna sensor for vital sign monitoring while the person is lying down in supine, prone, or lateral recumbent positions. For the supine and lateral recumbent positions, the signal quality is sufficiently high. However, in

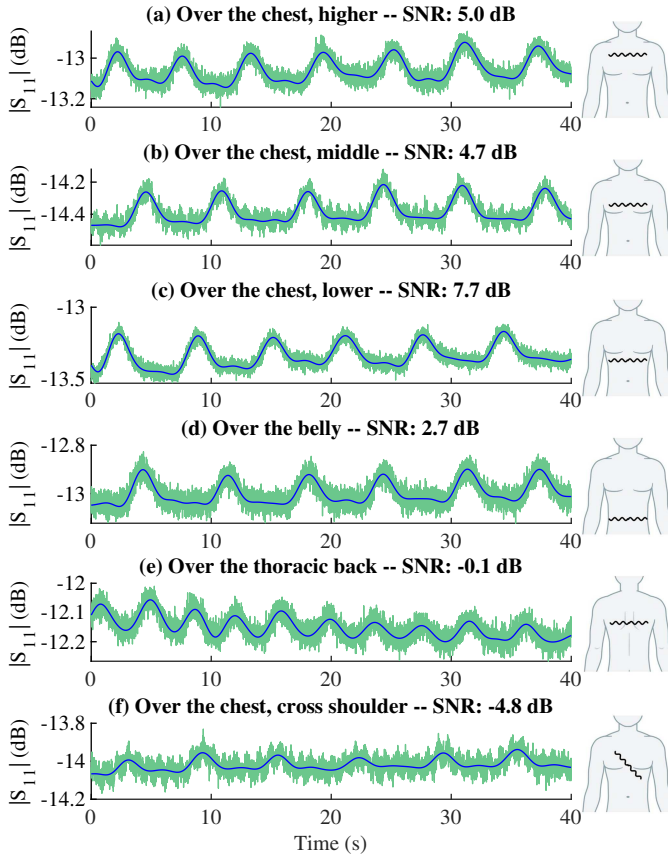


Fig. 2. Antenna performance while placed in different regions over the body. Blue trace shows the extracted respiratory signal.

the prone position, the signal quality drops. This degradation occurs because the antenna is sandwiched between the body and the mattress, with the added effect of metal springs within the mattress coming very close to the antenna. Despite the reduced signal-to-noise ratio (SNR), the respiration signal still remain detectable and also demonstrate a clear DFT in the 5–25 bpm respiratory frequency range for a 60 second window. This demonstrates that, although bringing metallic material close to the antenna can impact its performance, the signal can still be potentially extracted, provided the interrogation system has adequate sensitivity to capture the subtle signal.

C. Antenna Separation from the Chest

Antenna fields are strongest in the immediate vicinity of the antenna due to the presence of strong near-field components, as shown in Fig. 4. The near fields diminish rapidly as the observation point moves farther from the antenna and become negligible once the distance exceeds approximately half a wavelength. Therefore, to achieve the best effect on the antenna's impedance, it is preferable to place the antenna as close as possible to the chest surface, maximizing the interaction of the near fields with the internal organs being monitored. However, since the fields do not suddenly drop to zero, it is still possible to observe vital signs even if there is a gap between the chest surface and the antenna.

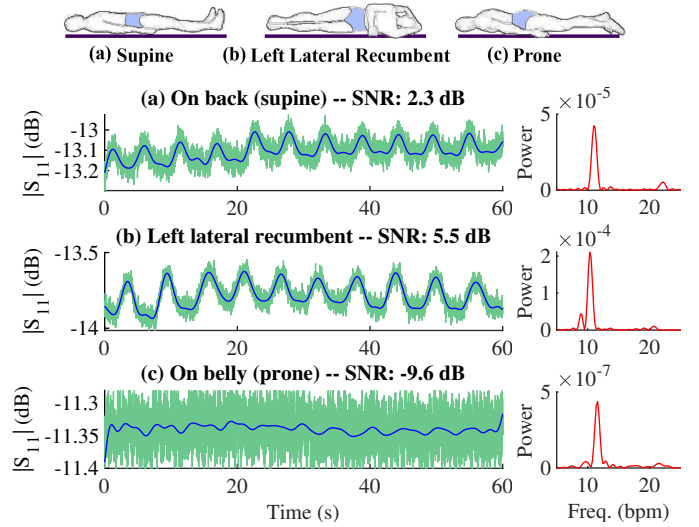


Fig. 3. Illustration of recumbent positions, corresponding antenna performances, and DFT in 5-25 bpm. Blue trace is the extracted respiratory signal.

To test this possibility, the antenna is installed in free space and placed at a distance d from the person's chest while they breathe, and the signals are recorded. The results of this experiment are shown in Fig. 5. From the results, it is evident that as the antenna is placed farther from the chest, the quality of the signal decreases. It can also be concluded that the antenna can be distanced from the chest surface up to one wavelength (λ) before the signal quality deteriorates to a point where it can no longer be reliably extracted. While a detector with very low noise might still be able to extract respiratory activity from a person standing at one wavelength away from the antenna, the chances of detecting cardiac activity diminish as the gap widens. This is because detecting cardiac activity requires significant penetration of the electrical fields into the body to reach the heart. However, chest wall movements due to respiration can still be detectable from a distance.

D. Antenna Frequency and Design

The SABioS method offers the flexibility of choosing different antenna types for capturing signals, along with the option to use either rigid or flexible antennas. Additionally, these antennas can be designed to operate at various frequencies, while still being effective for vital sign monitoring. A set of sample signals recorded using different types of antennas and operating frequencies is presented in Fig. 6. The samples

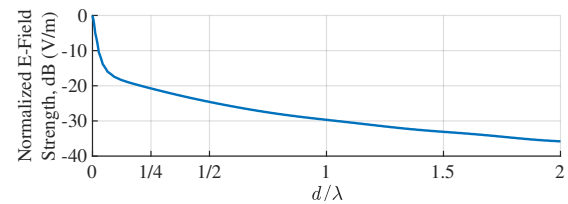


Fig. 4. Antenna E-field strength at distance d perpendicular to the center of the dipole, normalized to the E-field on the surface of the antenna.

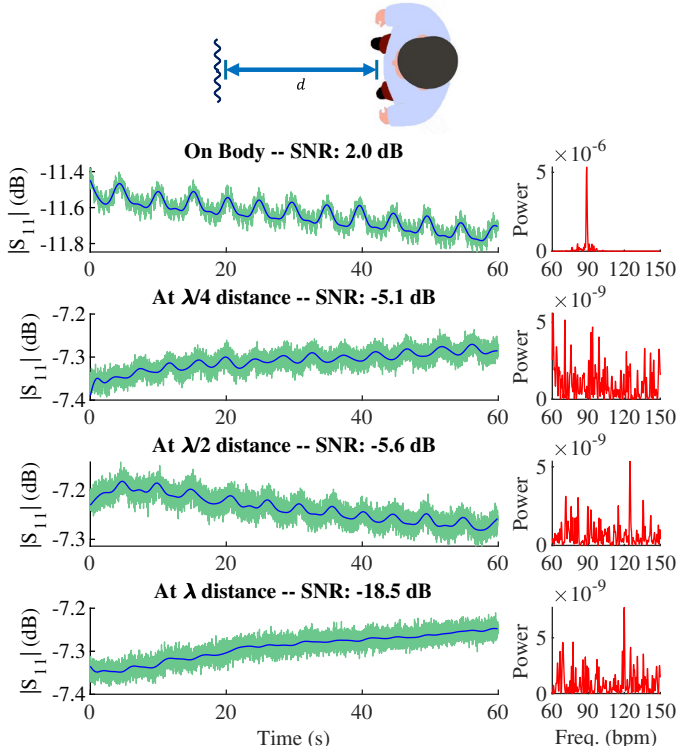


Fig. 5. Diagram showing the antenna in front of the subject in the distance d , the captured signals in different distances from the chest wall. The DFT in 60–150 bpm range shows that the heart rate estimation becomes more ambiguous as d increases. Blue trace is the extracted respiratory signal.

include data from four different types of antennas: sinusoidal dipoles, PCB monopoles, chip antennas, and microstrip patch antennas.

As shown in Fig. 6, SABioS can be implemented across different frequencies, with the experiments covering a range from 525 MHz to 2.8 GHz. The patch antenna operating at 2.8 GHz shows the lowest signal-to-noise ratio (SNR) and loses the harmonics associated with cardiac activity. This outcome is expected, as increasing the frequency reduces the wavelength, which in turn shortens the distance over which the intense near-field components dominate. This limits the reach of the antenna's near fields within the body, thus reducing its performance in detecting vital signs. In contrast, antennas operating at sub-gigahertz frequencies generally exhibit a good signal-to-noise ratio, with a clear cardiac signal visible. However, despite the larger near-field region and improved sensitivity to internal organ movements provided by larger antennas, the fabrication and integration of such antennas into wearable devices for vital sign monitoring can be challenging and may not be beneficial.

III. CONCLUSION

The application of SABioS for vital signs monitoring in various conditions was investigated, complementing our earlier study [13] where the method was examined in greater depth, validated against medical standards, and extensively compared with state-of-the-art. This work demonstrates that antennas

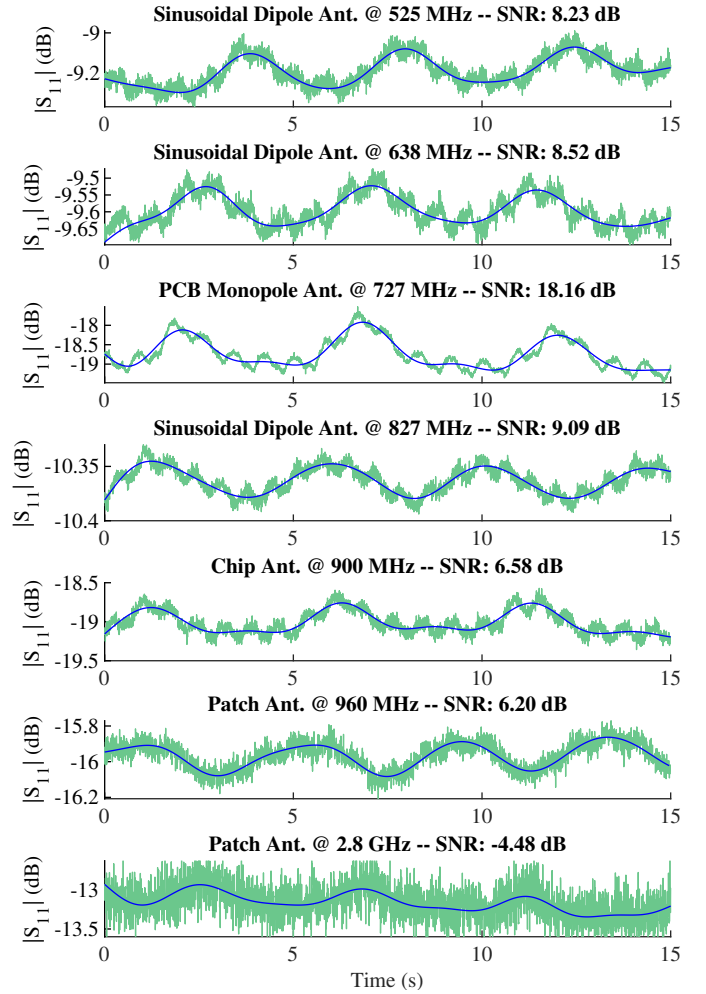


Fig. 6. Performance of SABioS method with different antennas operating in different frequencies

can be positioned at different locations on the upper body, depending on the targeted vital parameters and application. The method's performance was also evaluated in recumbent positions, showing its potential for in-bed monitoring. Additionally, its ability to detect vital signs from a distance was explored. Finally, the method's robustness across different frequencies and antenna types was assessed, highlighting its flexibility for deployment in diverse configurations.

While these experiments demonstrate the promising potential of SABioS for use in diverse applications, such as remote patient monitoring, managing chronic conditions like sleep apnea, or fatigue detection, its efficacy in each specific application requires further validation with larger participant groups. This includes comparing its performance with gold-standard medical devices, particularly regarding its capability to monitor cardiac activity and detect specific patterns such as breath cessation, hypo- or hyperventilation, and abnormal cardiac rhythms, among others. Additionally, its applicability in real-life scenarios for diagnosing or prognosticating various medical conditions should be evaluated.

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